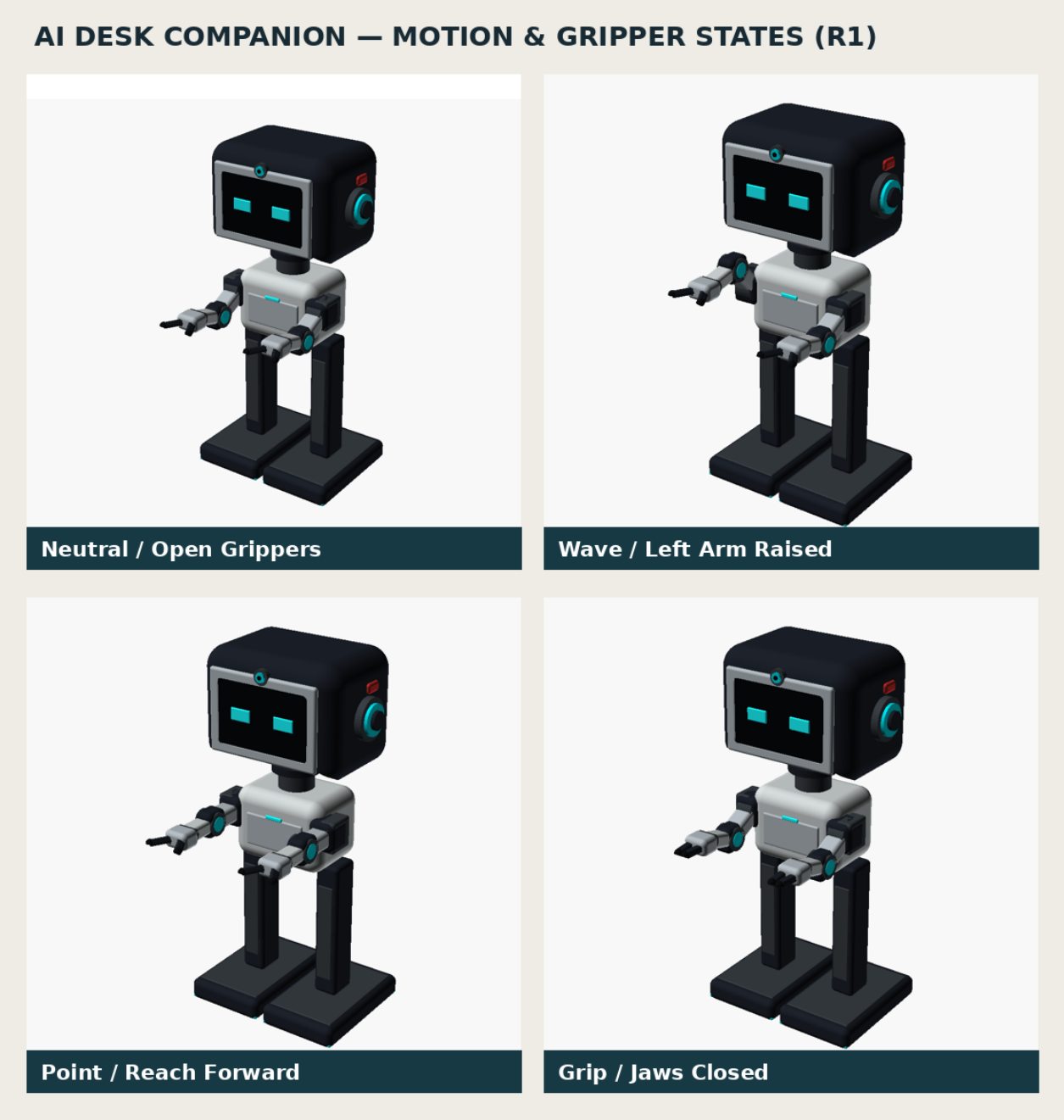
AI DESK COMPANION

CAD, Articulated Arm & Gripper Motion Report - Revision R1



Revision scope: Retains the selected two structural legs and broad feet; replaces the previous static arm shapes with separate shoulder links, elbow links, wrist/palm and open/closed pinch-gripper geometry. Includes discrete STEP poses and a parametric OpenSCAD animation source.

Important status: This is an articulated concept/packaging revision, not production-ready mechanism CAD. The animation demonstrates geometry movement, but does not prove collision-free travel, actuator torque, gripper force, reliability or safety.

1. What moves in R1

The current selected leg-and-foot design is a standing structure. The two legs are not powered, and the robot does not walk, roll or drive around a desk. The movement revision applies to the arms and hands only.

Motion	Shoulder pitch	Elbow relative	Gripper
Neutral	-25°	+40°	Open
Wave left/right	+45°	-55° on waving arm	Open
Point/reach	0°	0°	Open
Nudge pose	-20°	+55°	Open
Grip	-25°	+40°	Closed

Each pose is exported as its own STEP file. STEP normally contains static solids, so the model uses OpenSCAD for the live interpolated preview and STEP snapshots for CAD review.

2. Joint architecture

Joint	Motion axis	Proposed concept range	Count
Shoulder pitch	X axis; raises/lowers arm in Y-Z plane	-35° to +50°	1 per arm
Elbow pitch	X axis; relative angle to upper arm	-60° to +65°	1 per arm
Coupled gripper	Two jaw pivots around Z, one coupled actuator	Approx. 22.6 mm tip-center gap open; 5.6 mm closed	1 per hand
Head tilt	Not in R1 motion assembly	Requires hinge/dock redesign	0 in R1
Body locomotion	Not in R1	Structural legs/feet only	0 in R1

Provisional arm actuation count: six actuator channels total (two arm-joint actuators and one gripper actuator per arm). A future powered head-tilt joint adds at least one more actuator and requires its own mechanical design. The current docking interface should not be expected to tilt while remaining directly mated.

3. Hand/gripper design

Each hand is modeled as a palm block, two separate jaw/finger links, pivot-pin envelopes and soft rounded tips. The jaws move symmetrically in the OpenSCAD animation. The open tip-center spacing is about 22.6 mm; the closed tip-centers meet at about 5.6 mm, so the soft tips are nominally touching. With 2.8 mm radius tips, the approximate clear opening in the fully open state is 17 mm. These are geometry targets, not measured gripper capacity.

> The hand is intended for expressive gestures and very light pinching. It is not yet a load-rated gripper: the CAD does not yet define a final servo horn, coupling link dimensions, bearing/bushing clearances, retaining hardware, finger cut-outs, contact material or force limit. Do not claim it can safely lift objects until those items are designed and tested.

4. How to view movement

Live kinematic concept: open `motion_revision_R1/AI_Desk_Companion_Motion_Animated.scad` in OpenSCAD. Use View → Animate and set a convenient FPS/Steps value; the $t parameter drives neutral → wave → neutral → point → nudge → neutral. The gripper opening changes throughout the cycle. To hold a fixed point, set `animated = false` and change `motion_t` in the source.

Static CAD: use the separate full-assembly STEP poses for neutral/open, wave left, wave right, point, nudge, reach and closed-gripper states. The individual left/right arm STEP files provide arm-only subassemblies.

5. Hardware/control implications

A body controller such as an ESP32 can command the shoulder/elbow/gripper actuators, while the Raspberry Pi head sends high-level intents (for example WAVE_LEFT or GRIP_OPEN). The controller should translate each intent to validated joint targets, enforce per-axis bounds and speed limits, and reject overlapping or expired commands. The app should display the command state and allow cancellation.

Required protections include mechanical travel stops; conservative speed/current limits; safe stop on controller watchdog timeout or head/body link loss; no automatic motion restart after reconnect; and a hardware power cut for actuator power. Keep actuator power separate from the Pi logic rail and size the regulator/battery only after measured stall/transient current and thermal tests.

6. Verification status

Automated geometry check: 85 component solids in the R1 neutral assembly; 0 invalid individual solids reported by CadQuery. Nominal neutral assembly bounds are approximately 123.4 mm wide × 142.3 mm deep × 264.1 mm tall. These bounds are calculated from the current generated geometry.

What this check does not prove: solid intersections across every motion frame, correct hinge clearances, printable watertight parts, fastener engagement, structural stiffness, tipping margin, actuator torque, gripper force, battery runtime, thermal performance or physical safety.

[ ] Replace servo placeholders with exact actuator CAD and torque/current data.

[ ] Add bore geometry, bearing/bushing clearances, shafts/retainers and a final one-servo dual-jaw linkage.

[ ] Run a collision sweep over all allowed shoulder/elbow/jaw angles; restrict any colliding states.

[ ] Calculate worst-case centre of gravity and tipping margin with asymmetric arm poses.

[ ] Validate leg, foot, shoulder mount and head-dock loads for the selected materials/manufacturing process.

[ ] Measure current draw and enclosure temperatures during continuous arm motion and computer workloads.

[ ] Prototype the gripper at low force; test pinch points, stall detection, repeatability and stop behaviour.

[ ] Keep the legs stationary unless a separate powered mobility base is deliberately designed.

7. R1 files

AI_Desk_Companion_CAD_and_Motion_R1.zip contains the original base CAD/source, the articulated-arm revision, full-assembly STEP snapshots for named poses, left/right arm assemblies, the OpenSCAD motion source, joint/pose tables, the animated GIF and preview sheet. OpenSCAD animation and STEP snapshots describe conceptual movement; they are not motor firmware or manufacturing drawings.

8. Reference dimensions and hardware caveats

The camera/head remains an envelope model pending vendor CAD replacement. Raspberry Pi publishes the Pi 5 mechanical design files and recommends a 5 V / 5 A USB-C power supply for the recommended high-load configuration; Camera Module 3 Wide is listed at approximately 25 × 24 × 12.4 mm. These component references do not validate the complete head assembly.

Official references: Raspberry Pi 5 specifications; Camera Module 3 specifications.

Engineering status: R1 articulated motion concept — suitable for visual review and the next mechanical-design iteration; not cleared for manufacturing or unsupervised powered operation.